

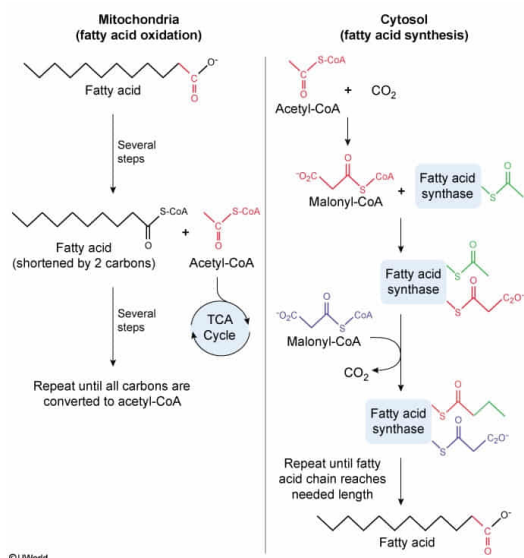
Fatty acid oxidation and synthesis occur in which subcellular compartments, respectively?

- ✓ ☒ A. In the mitochondria and the cytosol [77%]
☐ B. In the nucleus and the mitochondria
☐ C. In the cytosol and the nucleus [2%]
☐ D. In the cytosol and the mitochondria [19%]

Correct

76% answered correctly

Explanation:



Catabolic processes break down complex molecules for energy, whereas anabolic processes synthesize complex molecules from simple molecules for energy storage and for growth. Most catabolic pathways have opposing anabolic pathways. The enzymes that control **opposing processes** are tightly regulated so that only one pathway dominates at any given time. Regulation is typically achieved through a combination of:

- **Hormonal signaling**
- **Allosteric control**
- **Compartmentalization** of enzymes

Fatty acid oxidation is the catabolic breakdown of fatty acids to acetyl-CoA, whereas **fatty acid synthesis** is the anabolic production of fatty acids from acetyl-CoA units. These opposing pathways are regulated in part by compartmentalization. The enzymes required for **fatty acid oxidation** are found in the **mitochondria**, where newly generated acetyl-CoA molecules can enter the citric acid cycle and the electron transport chain to produce ATP. In contrast, the enzymes for **fatty acid synthesis** are in the **cytosol**, where excess acetyl-CoA is converted to malonyl-CoA and then linked to other acyl units to form fatty acids.

(Choices B and C) The primary metabolic process in the nucleus is the synthesis of nucleic acids (DNA and RNA) from nucleotides. Fatty acids are neither synthesized nor degraded in the nucleus.

(Choice D) This answer reverses the compartments in which fatty acid oxidation and synthesis occur.

Educational objective:

Opposing catabolic and anabolic pathways are regulated hormonally, allosterically, and by compartmentalization. Fatty acid oxidation (degradation) and fatty acid synthesis are compartmentalized, with oxidation occurring in the mitochondria and synthesis occurring in the cytosol.

Time spent:0 sec

Subject:
Biochemistry

QID:401893

System:
Metabolic Reactions